

NEXORA — Complete Setup & Deployment Guide

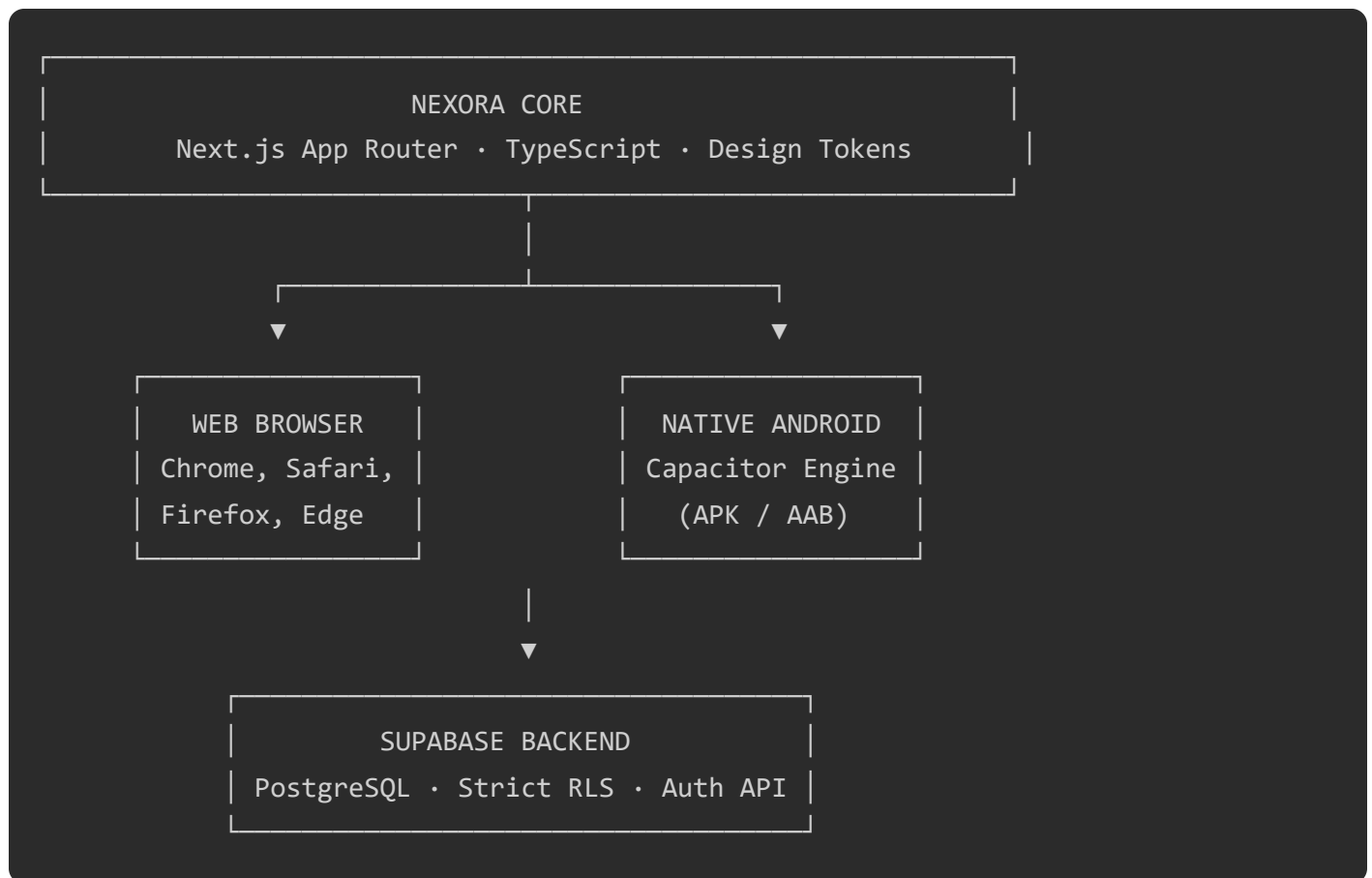
Work. Plan. Build. Live.

A unified work-and-life management platform.

One Codebase for Responsive Web & Native Android.

1. Overview & Architecture

NEXORA is engineered to run the **exact same codebase** across both **Web** (desktop, tablet, mobile browsers) and **Native Android** (APK / Google Play ready).



- **Frontend & App Logic:** Next.js (App Router), TypeScript, responsive CSS design tokens (`globals.css`), fully accessible touch targets ($\geq 44 \times 44 \text{pt}$).

- **Backend & Database:** PostgreSQL via Supabase with row-level multi-tenancy (`workspace_id`) and Row Level Security (RLS) enforced on 100% of tables.
 - **Mobile Engine:** Capacitor 8.x embedding the Next.js responsive app into native Android Studio project (`/android`), with zero code bifurcation.
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2. Prerequisites

Ensure you have the following installed on your development machine:

1. **Node.js:** Version 18.x, 20.x, or 22.x LTS ([Download Node.js](#)).
 2. **npm:** Version 9.x or higher (comes bundled with Node.js).
 3. **Git:** Installed and configured on your system.
 4. **Database (Supabase):**
 - **Option A (Recommended for Quick Start):** Free hosted project on [supabase.com](#).
 - **Option B (Local Dev):** Docker Desktop + Supabase CLI (`npx supabase start`).
 5. **For Android Development (Only needed when building/testing Android):**
 - [Android Studio](#) (Ladybug / Jellyfish or latest version).
 - Android SDK (API 33 or higher) & Command Line Tools.
 - Java Development Kit (JDK 17 or 21, bundled with Android Studio).
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3. Environment Variables Configuration

In the root of the project (`d:\nexora`), create your `.env.local` file:

```
# Copy the example file:
cp .env.local.example .env.local
```

Edit `.env.local` and provide your Supabase credentials:

```
# =====
# Supabase Configuration
# =====

# Public keys (Safe for browser & client bundles)
NEXT_PUBLIC_SUPABASE_URL=https://your-project-id.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=eyJhbGciOiI...your-anon-key
```

```
# Private Server-Only Key (§12.2: NEVER reaches client bundle or APK)
SUPABASE_SERVICE_ROLE_KEY=eyJhbGciOiI...your-service-role-key

# Application Base URL
NEXT_PUBLIC_APP_URL=http://localhost:3000
```

Where to find your Supabase credentials:

1. Go to your [Supabase Dashboard](#).
2. Select your project.
3. Click on **Project Settings** (gear icon) → **API**.
4. Copy **Project URL** → paste into **NEXT_PUBLIC_SUPABASE_URL**.
5. Copy **anon / public** key → paste into **NEXT_PUBLIC_SUPABASE_ANON_KEY**.
6. Copy **service_role / secret** key → paste into **SUPABASE_SERVICE_ROLE_KEY**.

4. Database Setup & Migrations

NEXORA comes with three forward-only, tested migrations in `/supabase/migrations/`:

- `20260903000001_foundation_tenancy.sql`: Workspaces, workspace members, teams, team members, RLS helper functions.
- `20260903000002_work_item_engine.sql`: Projects, work items, statuses, types, labels, comments, attachments, activity events, sequence generator.
- `20260903000003_rls_work_item_engine.sql`: Complete RLS policies for every table with mandatory `with check` clauses.

Applying to Hosted Supabase (Dashboard):

1. Open your Supabase Project Dashboard → **SQL Editor**.
2. Create a **New Query**.
3. Copy and run the contents of `supabase/migrations/20260903000001_foundation_tenancy.sql`.
4. Copy and run the contents of `supabase/migrations/20260903000002_work_item_engine.sql`.
5. Copy and run the contents of `supabase/migrations/20260903000003_rls_work_item_engine.sql`.

Applying via Supabase CLI (Alternative):

If you have linked your Supabase CLI (`npx supabase link --project-ref your-ref`):

```
npx supabase db push
```

5. Running the Web Application

Step 1: Install Dependencies

```
npm install
```

Step 2: Run Security & RLS Test Suite (\$15)

Before starting the app, verify that all 103 security tests pass:

```
npm run test:security
```

Expected output:

```
✓ tests/security/rls-coverage.test.ts (90 tests)
✓ tests/security/auth-rbac.test.ts (13 tests)
Test Files  2 passed (2)
Tests      103 passed (103)
```

Step 3: Run the Development Server

```
npm run dev
```

Open <http://localhost:3000> in your browser.

- Visit `/auth/signup` to create your initial user account.
- Visit `/onboarding` to create your first workspace.
- Visit `/onboarding/project` to create your first project and land directly on the interactive Kanban board!
- Press `c` anywhere to trigger the **Quick Create** modal.

Step 4: Build for Production

```
npm run build
npm start
```

6. Running on Android (Same Codebase)

NEXORA uses Capacitor to package the identical responsive application into a native Android application.

Step 1: Sync the Web Assets

Whenever you make updates to the web app:

```
npm run build
npm run android:sync
```

This syncs web assets and configurations into the native `android/` directory.

Step 2: Configure Live Development for Android

To test on an Android Emulator or physical phone connected to your Wi-Fi:

Open `capacitor.config.ts`:

```
import type { CapacitorConfig } from '@capacitor/cli';

const config: CapacitorConfig = {
  appId: 'com.nexora.app',
  appName: 'NEXORA',
  webDir: 'public',
  server: {
    androidScheme: 'https',
    cleartext: true,
    // FOR ANDROID EMULATOR:
    // url: 'http://10.0.2.2:3000',

    // FOR PHYSICAL ANDROID PHONE (Replace with your computer's Local Wi-Fi IP):
    // url: 'http://192.168.1.100:3000',
  },
};
```

```
// FOR PRODUCTION:
// url: 'https://your-nexora-app.vercel.app',
},
android: {
  backgroundColor: '#FAFBFC',
  allowMixedContent: true,
},
};

export default config;
```

Note for Android Emulator: `10.0.2.2` is Android's special alias to your computer's `localhost`. When running `npm run dev`, your emulator loads the live dev server directly with hot reload!

Step 3: Open in Android Studio

Run:

```
npm run android:open
```

Android Studio will open the `/android` project directory.

Step 4: Run on Device / Emulator

1. Wait for Android Studio to complete Gradle sync.
2. Select your device (or create an Android Virtual Device in Device Manager, e.g. Pixel 7 / Android 14+).
3. Click the green **Run** (▶) button in Android Studio.
4. NEXORA launches with native Android status bar, splash screen, touch interactions, and back-button support!

Step 5: Build a Standalone Android APK

To generate a release or debug `.apk` to install directly on phones without Android Studio:

On Windows (PowerShell):

```
cd android
.\gradlew.bat assembleDebug
```

The APK will be generated at:

```
android/app/build/outputs/apk/debug/app-debug.apk
```

You can transfer this APK directly to any Android device and install it.

7. Project Structure Reference (\$47)

```
d:/nexora/
├─ android/           # Native Android Studio project (Capacitor)
├─ app/               # Next.js App Router (UI & API Routes)
│   ├─ api/           # Server endpoints (projects, work-items, health)
│   ├─ auth/          # Authentication flows (login, signup, reset)
│   ├─ dashboard/     # Main dashboard & greeting
│   ├─ onboarding/    # Zero-config workspace & project creation
│   ├─ projects/[id]/ # Interactive Kanban board page
│   ├─ globals.css     # Token-based design system (light & dark)
│   └─ layout.tsx      # Root layout with accessibility & SEO
├─ components/
│   └─ board/          # KanbanBoard, WorkItemCard, QuickCreateModal, DetailD
├─ lib/
│   ├─ auth/           # RBAC engine & effective permission resolution
│   ├─ db/             # Typed database queries (work-items, workspaces, type
│   ├─ supabase/       # SSR and browser Supabase clients
│   ├─ validation/     # Server-side Zod validation schemas
│   └─ logger.ts       # Structured JSON logging with credential redaction
├─ supabase/
│   └─ migrations/    # Forward-only SQL migrations with complete RLS
├─ tests/
│   └─ security/       # Automated S1-S5 security & RLS test suite
├─ capacitor.config.ts # Android & cross-platform configuration
└─ package.json        # Scripts & dependencies
```

8. Verification & Troubleshooting

Scenario	Check / Solution
Security tests fail	Run <code>npm run test:security</code> . If modifying SQL, ensure every new table has <code>alter table <x> force row level security;</code> and all UPDATE policies have <code>with check (...)</code> .
Android cannot connect to localhost	Ensure <code>capacitor.config.ts</code> uses <code>http://10.0.2.2:3000</code> (for emulator) or your computer's LAN IP <code>http://192.168.x.x:3000</code> (for physical device) rather than <code>localhost</code> .
Missing Supabase Keys	Check <code>.env.local</code> . Remember that variables used in client code must start with <code>NEXT_PUBLIC_</code> .
Android build error	Run <code>npm run android:sync</code> before building in Android Studio. Ensure JDK 17+ is selected in <code>Android Studio > Settings > Build > Build Tools > Gradle > Gradle JDK</code> .

NEXORA is now fully set up and ready to run on Web and Android!